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Department of CSE (Artificial Intelligence & Machine Learning)

ROBUST AUDIO-VISUAL SPEECH SEPARATION SYSTEM

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INTRODUCTION

In real-world environments such as meetings, classrooms, and video conferencing platforms, multiple people often speak simultaneously. Traditional audio-only separation systems struggle in noisy and overlapping speech conditions.

Recent research in deep learning shows that visual cues such as lip motion can improve speech separation performance.

This project focuses on designing a robust audio-visual speech separation system that performs effectively even when the video input suffers from low frame rate or dropped frames

The system is designed to enhance speech clarity and support real-time caption generation for users with hearing impairments.

PROBLEM STATEMENT

In real-world video calls and recordings:

- Multiple speakers talk simultaneously
- Background noise corrupts audio
- Video frames may be dropped or delayed
- Visual input can be inconsistent

Existing audio-visual models assume high-quality synchronized video and fail under low-frame-rate conditions.

Therefore, there is a need to develop a robust audio-visual speech separation system that can dynamically adapt when video input is unreliable.

OBJECTIVES

- To study existing audio-visual speech separation techniques.
- To develop a multimodal deep learning model combining audio and visual features.
- To design a temporal attention mechanism for handling low-frame-rate video.
- To allow user-selected speaker extraction.
- To evaluate performance under noisy, corrupted and low-FPS conditions.
- To demonstrate application in assistive communication systems through enhanced speech clarity and caption generation support.

MOTIVATION

- Increasing use of video conferencing platforms
- Real-world video often suffers from lag and frame drops
- Existing systems work well only in controlled environments
- Practical need for robust speech separation
- Improved speech clarity and caption support can help people with hearing impairments.

LITERATURE SURVEY

Paper / Author	Key Methodology	Limitation
Looking to Listen *(Ephrat et al., 2018)* *(Base Paper)*	Uses a two-stream CNN to fuse audio and extracted face/lip features for separation.	**Assumes stable video quality.** Performance drops significantly if frames are missing or video lags.
The Conversation *(Afouras et al., 2018)*	Enhances both magnitude and phase of speech using deep audio-visual features.	**Computationally expensive.** Does not account for variable frame rates common in real-world calls.
Visual Voice *(Gao et al., 2021)*	Uses cross-modal embeddings and U-Net architecture for audio-visual target speaker separation.	Limited handling of **unreliable visual input**. Does not explicitly address low-FPS conditions.
Robust AV Speech *(Hong et al., 2019)*	Introduces a reliability scoring mechanism to adaptively weigh audio and video features.	Focused on **speech recognition** rather than generating clean separated speech signals.
Audio-Visual Feature Synchronization *(Saleem et al., 2025)*	Aligns audio and visual features to enhance speech clarity in noisy hearing-aid environments.	Designed for **speech enhancement**, not multi-speaker separation.

NOVELTY

EXISTING SYSTEM

- Uses only audio to separate voices
- Some systems use both audio and video
- Assumes video is clear and synchronized
- Performance drops when video frames are missing
- Cannot adapt to low frame rate
- Not fully suitable for real-world video calls

PROPOSED SYSTEM

- Uses audio and visual (lip motion) features for speech separation
- Introduces temporal attention for dynamic audio-visual fusion
- Adapts automatically when visual frames are unreliable
- Maintains performance under low frame rate and dropped frames
- Allows target speaker selection from multi-speaker video
- Generates clean separated speech output
- Supports hearing-impaired users through clearer speech and captions.

SCOPE

- Application in video conferencing platforms
- Assistive support for people with hearing impairments
- Speech enhancement in surveillance systems
- Real-world noisy communication environments like classrooms and meetings
- Integration with real-time caption systems

CHALLENGES

- Synchronization of audio and video streams
- Handling different frame rates
- Training the model
- Dataset preprocessing

PO'S, PSO'S AND SDG'S

PO's:

- Engineering knowledge & Problem analysis (PO1, PO2)
- Design and development of solutions using modern tools (PO3, PO5)

PSO's:

- Application of AI & ML in real-world systems (PSO1)
- Development of intelligent multimedia systems (PSO2)

SDG:

- SDG 9: Industry, Innovation and Infrastructure
- SDG 4: Quality Education (assistive communication support)

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THANK YOU